

DD_SUB-WF-TERMINATE-01-FLOW-WIK - WIK Termination BPMN

Developer implementation guide: DD_SUB-WF-TERMINATE-01-FLOW-WIK-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

This DD defines the WIK operator BPMN for subscription termination.

Process key:

```
sub-terminate-wik
```

This BPMN implements the parent termination contract from DD_SUB-WF-TERMINATE-01-Subscription_Termination-v1.0.md for WIK.

The WIK termination flow is:

```
validate -> cancel in-flight operation -> PENDING_TERMINATION -> compute billing window -> settlement -> resolve equipment
```

2. Ownership

This DD owns:

- WIK-specific termination BPMN process key.
- WIK happy-path BPMN sequence.
- WIK cooling-off supervisor task behavior.
- WIK equipment disposition behavior.
- WIK service revoke and HomePass release sequencing.
- WIK worker-topic composition.
- WIK message catches.
- WIK use of rules.subscription.terminate.wik.
- WIK termination examples.

This DD does not own:

- Generic termination API contract and rules. See parent termination DD.
- Shared operation framework. See SUB-WF-FRAMEWORK-01.
- Worker contracts. See DD_SUB-WF-FRAMEWORK-01-WORKERS.
- Worker implementation code.
- SUB-LM, BIL, FUL, NOT, OSR-RMA, BIL-04, scheduler internals.

3. Required Runtime Configuration

3.1 Framework Operation Config

What this table is for:

subscription_operation_config tells the subscription workflow framework which BPMN process to start for an operator and operation kind.

When to use it:

Use it when the API receives a TERMINATE request and needs to resolve the WIK BPMN process key.

Required row:

```
INSERT INTO subscription_operation_config (  
  operator_code,  
  operation_kind,  
  default_bpmn_process_key,  
  operation_timeout_seconds,
```

```

        feature_flags,
        enabled,
        updated_by
    )
VALUES (
    'WIK',
    'TERMINATE',
    'sub-terminate-wik',
    120,
    '{"customer_self_service_enabled": true, "cooling_off_supervisor_override_required": true, "churn_save_attempt_enabled": true}',
    TRUE,
    'migration'
);

```

Column meaning:

Column	Purpose	Sample	Meaning
operator_code	Operator scope.	WIK	This row applies to WIK.
operation_kind	Subscription operation.	TERMINATE	This row is used for termination requests.
default_bpmn_process_key	Camunda process key to start.	sub-terminate-wik	Start the WIK termination BPMN.
operation_timeout_seconds	Framework operation timeout.	120	WIK termination should finish quickly unless waiting on supervisor or fulfillment.
feature_flags	WIK-specific operation flags.	{"cooling_off_supervisor_override_required":true}	Supervisor task is enabled; churn-save is not in v1.0.
enabled	Runtime enablement.	TRUE	API may start WIK termination.
updated_by	Audit actor.	migration	Seeded by migration.

3.2 WIK Termination Config

What this table is for:

subscription_termination_config stores WIK termination policy.

When to use it:

Load it before Drools validation and pass it as a fact into rules.subscription.terminate.wik.

Required row:

```

INSERT INTO subscription_termination_config (
    operator_code,
    cooling_off_window_days,
    default_equipment_disposition,
    dunning_equipment_disposition,
    dunning_auto_write_off_threshold,
    dunning_auto_write_off_currency,
    customer_self_service_enabled,
    cooling_off_supervisor_override,
    churn_save_attempt_enabled,
    updated_by
)
VALUES (
    'WIK',
    7,
    'PENDING_COLLECTION',
    'PENDING_COLLECTION',
    10000.00,
    'KES',
    TRUE,
    TRUE,
    FALSE,
    'migration'
);

```

Column meaning:

Column	Purpose	Sample	Meaning
operator_code	Operator scope.	WIK	Policy applies to WIK.
cooling_off_window_days	Cooling-off period.	7	Customer-requested termination inside 7 days is cooling-off.
default_equipment_disposition	Default non-dunning equipment handling.	PENDING_COLLECTION	Collect WIK equipment by default.

Column	Purpose	Sample	Meaning
dunning_equipment_disposition	Default dunning equipment handling.	PENDING_COLLECTION	Collect equipment after dunning termination unless write-off applies.
dunning_auto_write_off_threshold	Threshold for dunning write-off policy.	10000.00	BIL-01 may return write-off marker around this policy.
dunning_auto_write_off_currency	Threshold currency.	KES	Threshold is in Kenyan shillings.
customer_self_service_enabled	Allows customer portal termination.	TRUE	WIK customers may request termination through self-service.
cooling_off_supervisor_override	Enables supervisor task.	TRUE	WIK cooling-off cases require supervisor decision.
churn_save_attempt_enabled	Enables churn-save task.	FALSE	No WIK churn-save BPMN step in v1.0.

4. BPMN Start Variables

The shared framework starts Camunda with these variables.

Variable	Purpose	Example
operationId	Links BPMN to subscription_operation.	subop_01HZ_KAMAU_TERM
operatorCode	Operator scope.	WIK
subscriptionId	Subscription being terminated.	sub_01HZ_KAMAU_FIBER
operationKind	Always TERMINATE.	TERMINATE
customerId	Customer reference from SUB-LM.	cust_01HZ_KAMAU
homepassId	HomePass to release.	hp_01HZ_KAREN_017
terminationTrigger	Trigger requesting termination.	CUSTOMER_REQUESTED
terminationReasonCode	Operator/CVM reason.	CUSTOMER_MOVED_NO_COVERAGE
resolvedEquipmentDisposition	Final equipment disposition.	PENDING_COLLECTION
coolingOffApplicable	Whether cooling-off applies.	false
requiresCoolingOffSupervisorDecision	Whether supervisor user task runs.	false
equipmentBindings	Equipment at customer premises.	[{"equipmentRef": "ont_01HZ"}]
initiatingActorUserId	Actor user/service.	cc-agent-wik-007
initiatingActorRole	Actor role.	CALL_CENTER
initiatingWorkflowKind	Upstream workflow.	BQ_UI
initiatingWorkflowRef	Upstream reference.	cc-ticket-2026-07-20-4421

5. Worker Topics Used

These are logical Camunda external-task topics, not separate deployments. One worker app may subscribe to many topics, but logs, retries, metrics, and contracts must remain visible per topic.

Worker topic	Why WIK termination uses it
drools-decide	Re-check eligibility, role, cooling-off, and equipment disposition.
sub-lm-cancel-in-flight-operation	Cancel in-flight lifecycle operation before terminal flow.
sub-lm-put-pending-status	Move subscription to PENDING_TERMINATION.
sub-lm-compute-cycle-window	Compute billing-cycle context for BIL-01.
bil01-emit-intent	Ask BIL-01 to compute TERMINATION_INTENT settlement.
ful-call-terminate	Ask FUL-05 to revoke service.
ful-call-release-homepass	Release HomePass through FUL-07 or approved stub.
sub-lm-put-master-fields	Write termination master fields.
sub-lm-commit-state	Commit subscription to TERMINATED.
kafka-emit	Write termination success/failure event to outbox.
notification-send	Send termination notification without failing BPMN on notification failure.
subscription-operation-finalize	Set operation terminal state.

Worker topic	Why WIK termination uses it
sub-lm-revert-to-prior	Compensation before terminal commit, when safe.

6. Happy Path BPMN

```

Start
-> update operation state VALIDATING
-> drools-decide
-> gateway: eligible?
    no -> termination rejected path
    yes
-> gateway: in-flight operation exists?
    yes -> sub-lm-cancel-in-flight-operation -> wait cancellation acknowledgement
    no -> continue
-> update operation state PENDING_STATE_FLIP
-> sub-lm-put-pending-status
-> sub-lm-compute-cycle-window
-> update operation state BILLING_CALL
-> bil01-emit-intent
-> drools-decide resolve-equipment-disposition
-> gateway: cooling-off supervisor required?
    yes -> user task cooling-off-override-decision
    no -> continue
-> update operation state FULFILLMENT_CALL
-> ful-call-terminate
-> wait fulfillment-completed
-> ful-call-release-homepass
-> update operation state COMMITTING_FINAL_STATE
-> sub-lm-put-master-fields
-> sub-lm-commit-state
-> update operation state EMITTING_EVENT
-> kafka-emit SubscriptionTerminated
-> notification-send
-> subscription-operation-finalize COMPLETED
-> End

```

7. Step Details

7.1 drools-decide - Validation And Cooling-Off

Inputs:

```

{
  "kjarPackage": "rules.subscription.terminate.wik",
  "requestedDecisions": [
    "validate-eligibility",
    "compute-cooling-off-applicability"
  ],
  "factsJson": "{...termination facts...}"
}

```

Expected output for Kamau:

```

{
  "decisionsJson": {
    "eligible": true,
    "coolingOffApplicable": false,
    "daysActiveInContract": 920,
    "requiresCoolingOffSupervisorDecision": false
  }
}

```

If eligible = false, route to termination rejection.

7.2 sub-lm-put-pending-status

Inputs:

```

{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "toStatus": "PENDING_TERMINATION",
  "currentTransitionType": "TERMINATE",
  "currentTransitionReasonCode": "CUSTOMER_MOVED_NO_COVERAGE"
}

```

Output consumed later:

```

{

```

```

    "priorStatusSnapshot": {
      "statusCode": "ACTIVE",
      "currentTransitionType": null,
      "currentTransitionReasonCode": null
    }
  }
}

```

7.3 bil01-emit-intent

Inputs:

```

{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "intentKind": "TERMINATION_INTENT",
  "intentContext": {
    "terminationTrigger": "CUSTOMER_REQUESTED",
    "terminationReasonCode": "CUSTOMER_MOVED_NO_COVERAGE",
    "priorStatus": "ACTIVE",
    "coolingOffApplicable": false,
    "daysActiveInContract": 920,
    "contractMinimumTermMonths": 12,
    "equipmentDispositionRequested": "PENDING_COLLECTION"
  },
  "cycleWindow": {
    "cycleStart": "2026-07-01",
    "cycleEnd": "2026-07-31"
  }
}

```

Expected result:

```

{
  "intentDecisionId": "bidec_01HZ_KAMAU_TERM",
  "settlement": {
    "lineItems": [
      {
        "kind": "PRORATED_REFUND",
        "amount": -1200.00,
        "currency": "KES"
      }
    ],
    "totals": {
      "net": -1200.00,
      "currency": "KES"
    },
    "refundAmount": 1200.00,
    "writeOffMarker": false
  }
}

```

7.4 drools-decide - Resolve Equipment Disposition

Inputs:

```

{
  "kjarPackage": "rules.subscription.terminate.wik",
  "requestedDecisions": [
    "resolve-equipment-disposition"
  ],
  "factsJson": "{caller disposition, WIK config, billing writeOffMarker}"
}

```

Expected output:

```

{
  "equipmentDisposition": "PENDING_COLLECTION"
}

```

Caller-supplied disposition wins unless policy explicitly rejects it. For dunning termination with no supplied disposition, BIL-01 writeOffMarker=true may resolve to WRITTEN_OFF.

7.5 Cooling-Off Supervisor User Task

The user task runs only when:

```

terminationTrigger == CUSTOMER_REQUESTED
AND coolingOffApplicable == true
AND cooling_off_supervisor_override == true

```

User task:

```

cooling-off-override-decision

```

Candidate group:

COOLING_OFF_SUPERVISORS

Form fields:

Field	Purpose
overrideApproved	Supervisor approves continuing termination.
overrideReason	Required reason when approval is true.
notes	Optional supervisor note.

If not completed within 24 hours, alert operations through ICN-01. Do not auto-terminate without the configured decision.

7.6 ful-call-terminate

Inputs:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "intent": "TERMINATE_SERVICE",
  "reasonCategory": "CUSTOMER_REQUESTED",
  "equipmentDisposition": "PENDING_COLLECTION",
  "equipmentBindings": [
    {
      "equipmentRequirement": "GPON_ONT",
      "equipmentRef": "ont_01HZ_KAMAU"
    },
    {
      "equipmentRequirement": "ATA",
      "equipmentRef": "ata_01HZ_KAMAU"
    }
  ],
  "fulfillmentCorrelationId": "subop_01HZ_KAMAU_TERM:terminate"
}
```

FUL-05 revokes service, for example voice profile deletion and OLT/AAA deprovisioning. The BPMN waits for fulfillment-completed or fulfillment-failed correlated by fulfillmentCorrelationId.

7.7 ful-call-release-homepass

Inputs:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "homepassId": "hp_01HZ_KAREN_017",
  "releaseReason": "SUBSCRIPTION_TERMINATED",
  "fulfillmentCorrelationId": "subop_01HZ_KAMAU_TERM:release-homepass"
}
```

Until FUL-07 ships, this worker may call an approved stub. The stub must persist a durable release request and emit metrics. If no durable stub exists, route failures to manual recovery before committing TERMINATED.

7.8 sub-lm-put-master-fields

Inputs:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "fieldsJson": {
    "terminated_at": "2026-07-20T14:30:18Z",
    "termination_trigger": "CUSTOMER_REQUESTED",
    "termination_reason_code": "CUSTOMER_MOVED_NO_COVERAGE",
    "prior_status_at_termination": "ACTIVE",
    "last_status_changed_at": "2026-07-20T14:30:18Z"
  }
}
```

This step writes audit fields before the final state commit.

7.9 sub-lm-commit-state

Inputs:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "expectedFromStatus": "PENDING_TERMINATION",
  "toStatus": "TERMINATED"
}
```

```
}
```

Emit success only after this commit succeeds.

8. Message Catches

Message	Correlation key	Used when
fulfillment-completed	fulfillmentCorrelationId	FUL-05 revoked service, and optionally HomePass release completed.
fulfillment-failed	fulfillmentCorrelationId	FUL-05 or HomePass release failed.

Use distinct correlation IDs:

```
subop_01HZ_KAMAU_TERM:terminate
subop_01HZ_KAMAU_TERM:release-homepass
```

9. Drools Package

Required package:

```
rules.subscription.terminate.wik
```

Required decisions:

Decision	Purpose
validate-eligibility	Checks not terminated, trigger, role, and self-service config.
compute-cooling-off-applicability	Computes cooling-off and active days.
resolve-equipment-disposition	Resolves caller-supplied, default, dunning default, or write-off disposition.

Example rule intent:

```
rule "WIK termination rejects already terminated"
when
    $req: TerminateRequestFact()
    $sub: SubscriptionStateFact(subscriptionId == $req.subscriptionId, statusCode == "TERMINATED")
then
    modify($req) {
        setEligible(false);
        setEligibilityReasonCode("ALREADY_TERMINATED")
    }
end

rule "WIK customer requested termination allowed roles"
when
    $req: TerminateRequestFact(
        terminationTrigger == "CUSTOMER_REQUESTED",
        eligible == null,
        initiatingActorRole not in ("CUSTOMER_SELF_SERVICE", "CALL_CENTER", "SUBSCRIPTION_OPERATOR")
    )
then
    modify($req) {
        setEligible(false);
        setEligibilityReasonCode("ROLE_NOT_AUTHORIZED_FOR_TRIGGER")
    }
end

rule "WIK dunning write-off disposition"
salience 10
when
    $req: TerminateRequestFact(terminationTrigger == "DUNNING_TERMINATION", equipmentDispositionRequested == null)
    $intent: BillingIntentDecisionFact(writeOffMarker == true)
then
    modify($req) {
        setEquipmentDisposition("WRITTEN_OFF")
    }
end
```

10. Compensation Paths

Failure point	BPMN compensation
Drools rejects request	Emit SubscriptionTerminationRejected; finalize FAILED.
In-flight cancellation fails	Do not start termination; return OPERATION_INTERRUPTION_FAILED.

Failure point	BPMN compensation
Pending status update fails	No committed mutation expected; emit rejected; finalize FAILED.
Billing settlement fails	Run sub-lm-revert-to-prior; emit rejected; finalize FAILED.
FUL-05 revoke fails before network change	Run sub-lm-revert-to-prior; emit rejected; finalize FAILED.
FUL-05 partial/unknown revoke	Manual recovery.
HomePass release fails	Retry or manual recovery unless a durable release stub is accepted.
Master fields or state commit fails	Manual recovery.
Kafka emit fails	Outbox retries. BPMN should not mark event delivered synchronously.
Notification fails	Log and continue. Notification is reliability-isolated.

11. Worked Example

Kamau calls WIK Call Center to terminate because he is moving to a non-covered area.

Request:

```
POST /api/subscriptions/sub_01HZ_KAMAU_FIBER/terminate
Idempotency-Key: term-sub_01HZ_KAMAU-20260720T143000-cc007
Content-Type: application/json

{
  "operatorCode": "WIK",
  "terminationTrigger": "CUSTOMER_REQUESTED",
  "terminationReasonCode": "CUSTOMER_MOVED_NO_COVERAGE",
  "equipmentDisposition": "PENDING_COLLECTION",
  "customerNotes": "Moving to a non-covered area; no service available at new address",
  "initiatingWorkflowKind": "BO_UI",
  "initiatingWorkflowRef": "cc-ticket-2026-07-20-4421"
}
```

Runtime result:

```
API validates request
-> loads SUB-LM subscription summary: ACTIVE
-> loads WIK termination config and equipment bindings
-> Drools accepts CALL_CENTER CUSTOMER_REQUESTED termination
-> framework creates operation subop_01HZ_KAMAU_TERM
-> Camunda starts sub-terminate-wik
-> subscription moves ACTIVE -> PENDING_TERMINATION
-> billing receives TERMINATION_INTENT and computes prorated refund
-> Drools resolves equipmentDisposition=PENDING_COLLECTION
-> cooling-off supervisor task is skipped because subscription is past cooling-off window
-> FUL-05 revokes service
-> HomePass release worker releases/stubs release
-> termination master fields are written
-> subscription moves PENDING_TERMINATION -> TERMINATED
-> SubscriptionTerminated event is emitted
-> notification is sent
-> operation is COMPLETED
```

After success:

- Subscription status is TERMINATED.
- Customer service is revoked.
- HomePass slot is released or has a durable release request.
- SubscriptionTerminated includes settlement and equipment bindings.
- OSR-RMA creates pickup work order because equipment disposition is PENDING_COLLECTION.

12. Cross-DD Dependencies

Dependency	Why it matters
DD_SUB-WF-TERMINATE-01-Subscription_Termination-v1.0.md	Owns generic termination API, trigger rules, events, and invariants.
DD_SUB-WF-FRAMEWORK-01-Subscription_Workflow_Framework-v1.0.md	Owns operation creation, interrupt handling, and process start.
DD_SUB-WF-FRAMEWORK-01-WORKERS-v1.0.md	Owns worker-topic contracts.
DD_SUB-LM-01-Subscription_Lifecycle_Management-v1.0.md	Owns subscription status source of truth.

Dependency	Why it matters
BIL -01	Owns termination settlement.
BIL -04	Calls dunning termination and consumes termination event.
FUL -05	Owns network/service revocation.
FUL -07	Owns HomePass release; stub is temporary.
OSR-RMA-01 EQP	Creates pickup work order from termination event.
NOT -01	Owns customer notification.

13. Open Items

- FUL-07 HomePass release DD is still pending; WIK uses approved release stub until then.
- WUG, WTZ, and YASSN termination flow satellites still require operator-specific deep dives.